

PAIHENG XU

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RESEARCH INTEREST

Natural Language Processing, Computational Social Science, AI for Education

EDUCATION

- | | |
|---|-----------------------|
| University of Maryland, College Park, USA
Ph.D. Student in Computer Science | <i>2021 - present</i> |
| Johns Hopkins University, Baltimore, USA
M.S.E in Computer Science | <i>2018 - 2020</i> |
| Southwest University, Chongqing, China
B.E in Computer Science | <i>2014 - 2018</i> |

RESEARCH EXPERIENCE

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| UNIVERSITY OF MARYLAND, COLLEGE PARK
Research Assistant – Advisor: Wei Ai & Jing Liu | College Park, MD
<i>Sep. 2021 - present</i> |
| <ul style="list-style-type: none">· Hypothesis generation for text analysis by combining Sparse Autoencoder with causality-inspired frameworks.· Investigate bias and spurious correlation mitigation for text classification and LLM alignment.· AI for Edu - LLM-based transcript analysis to identify effective teaching practices and measure teaching quality. | |
| Research Assistant – Advisor: Louiqa Raschid & Vanessa Frias-Martinez | <i>Jan. 2023 - present</i> |
| <ul style="list-style-type: none">· Model correlations between geo-co-location and social media engagement using statistical & LLM-based methods.· Link online interaction patterns with offline community outcomes to promote pro-social behaviors. | |
| JOHNS HOPKINS UNIVERSITY
Full-time Research Developer – Advisor: Mark Dredze | Baltimore, MD
<i>Jun. 2020 - Aug. 2021</i> |
| <ul style="list-style-type: none">· Develop Twitter-based social mobility indices to study demographic differences in social distancing behavior. [link]· Apply embedding- and transformer-based models to discover e-cig brands and track sentiment shifts on social media. | |
| Research Assistant – Mentor: Zach Wood-Doughty | <i>Nov. 2018 - May 2020</i> |
| <ul style="list-style-type: none">· Demographic inference on Twitter with distant supervision (Package); study linguistic variation across racial groups. | |

INDUSTRY EXPERIENCE

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| ADOBE RESEARCH
Research Scientist Intern – Mentor: Haoliang Wang | San Jose, CA
<i>May 2025 - Aug. 2025</i> |
| <ul style="list-style-type: none">· Design style understanding and personalized suggestion generation through LVLM-based design pattern uncovering.· User-centric evaluation for personalization using paired image comparisons to capture design preferences. | |
| ADOBE RESEARCH
Research Scientist Intern – Mentor: Vishy Swaminathan, Gang Wu, & Xiang Chen | San Jose, CA
<i>May 2024 - Nov. 2024</i> |
| <ul style="list-style-type: none">· Skill development for software scripting through code generation & offline simulations using LLM-based agents. | |
| ADOBE RESEARCH
Research Scientist Intern – Mentor: Tong Yu & Haoliang Wang | Remote
<i>May 2022 - Aug. 2022</i> |
| <ul style="list-style-type: none">· Learning knowledge from Critical Service Outage reports for auto-remediation.· Adaptive Question Generation for Extractive Question Answering using In-context Learning. | |

SELECTED PUBLICATIONS

* denotes equal contribution

Paiheng Xu, Jing Liu, Wei Ai, Conditional Hypothesis Generation for LLM-Based Text Analysis with Researcher-Specified Covariates. *arXiv:2606.03029*, 2026

Nishant Balepur, **Paiheng Xu**, Wei Ai, Eunsol Choi, Rachel Rudinger, Jordan Lee Boyd-Graber, Check The Scoreboard: An Analysis of Scoring Schemes on Multiple-Choice Evaluation. *Under review*.

Paiheng Xu, Louiqa Raschid, Vanessa Frias-Martinez, Does Geo-co-location Matter? A Case Study of Public Health Conversations during COVID-19. *ICWSM 2026*.

Paiheng Xu, Gang Wu, Xiang Chen, Tong Yu, Chang Xiao, Franck Dernoncourt, Tianyi Zhou, Wei Ai, Vishy Swaminathan, Skill Discovery for Software Scripting Automation via Offline Simulations with LLMs. *Findings of EACL*, 2026.

Paiheng Xu*, Xiaoyu Liu*, Junda Wu, Jiaxin Yuan, Yifan Yang, Yuhang Zhou, Fuxiao Liu, Tianrui Guan, Hao-liang Wang, Tong Yu, Julian McAuley, Wei Ai, Furong Huang, Large Language Models and Causal Inference in Collaboration: A Comprehensive Survey. *Findings of NAACL 2025*.

Paiheng Xu*, Yuhang Zhou*, Bang An, Wei Ai, Furong Huang, GFairHint: Improving Individual Fairness for Graph Neural Networks via Fairness Hint. *Transactions on Knowledge Discovery from Data*, 2025

Paiheng Xu, Jing Liu, Nathan Jones, Julie Cohen, Wei Ai, The Promises and Pitfalls of Using Language Models to Measure Instruction Quality in Education. *NAACL 2024*.

Yuhang Zhou, **Paiheng Xu**, Xiaoyu Liu, Bang An, Wei Ai, Furong Huang, Explore Spurious Correlations at the Concept Level in Language Models for Text Classification. *ACL 2024*.

Paiheng Xu, David A. Broniatowski, Mark Dredze, Twitter Social Mobility Data Reveal Demographic Variations in Social Distancing Practices During the COVID-19 Pandemic. *Scientific Reports*, 2024.

Yuhang Zhou*, Jing Zhu*, **Paiheng Xu**, Xiaoyu Liu, Xiyao Wang, Danai Koutra, Wei Ai, Furong Huang, Multi-Stage Balanced Distillation: Addressing Long-Tail Challenges in Sequence-Level Knowledge Distillation. *Findings of EMNLP 2024*.

Paiheng Xu*, Zach Wood-Doughty*, Xiao Liu, Mark Dredze, Using Noisy Self-Reports to Predict Twitter User Demographics. *SocialNLP@NAACL*, 2021.

Paiheng Xu, Mark Dredze, David A. Broniatowski, The Twitter Social Mobility Index: Measuring Social Distancing Practices from Geolocated Tweets. *Journal of Medical Internet Research*, 2020.

Paiheng Xu, Likang Yin, Zhongtao Yue, Tao Zhou, On Predictability of Time Series. *Physica A: Statistical Mechanics, its Applications* 523, 345-351 (2019).

TECHNICAL SKILLS

Languages	Python, MATLAB, R, C/C++, C#, L ^A T _E X, Markdown, Bash
Tools	Transformers, PyTorch, TRL, HF Agents, Scikit-learn, NLTK, Statsmodels, NetworkX

HONOR & AWARDS

Dean's Fellowship, University of Maryland
National Scholarship, Ministry of Education, China
Outstanding Graduates, Southwest University

TEACHING & SERVICE

Teaching Assistant: Introduction to Artificial Intelligence, Fall 2022

Program Committee: ICWSM 2022, 2025-26

Reviewer: ARR 2025-26; KDD 2025; TheWebConf 2025; TPAMI

Sub-Reviewer: ACL 2021; KDD 2022; SIGIR 2022-2025